

A quadratic-grid barrier for whole-copy insertion

Research notes, 30 September 2026

Status. Complete proofs below. This is a barrier for a specified construction framework, not a lower bound for arbitrary fair dice. The quadratic scale of positive equidistant quadrature is classical; the purpose here is a self-contained explicit bound and its consequence for the dice insertion scheme.

Theorem 1 (Positive quadrature requires a quadratic grid). *Let $r \geq 2$ and suppose non-negative weights on a subset of $\{0, 1/N, \dots, 1\}$ integrate every polynomial of degree at most r exactly for the uniform probability measure on $[0, 1]$. Put $m = \lfloor (r - 2)/2 \rfloor$. Then*

$$N \geq \frac{m^2 + 4m + 6}{4} \geq \frac{r^2}{16}.$$

The weights may be arbitrary real numbers; they need not be equal, rational, or positive at every grid point.

Proof. Let $q(x) = P_m^{(0,3)}(2x - 1)$, using the standard Jacobi normalization. The following two norm identities are proved below:

$$A_m := \int_0^1 x q(x)^2 dx = \frac{m^2 + 4m + 6}{12}, \quad B_m := \int_0^1 x^2 q(x)^2 dx = \frac{1}{3}.$$

The degree of $f(x) = x(1/N - x)q(x)^2$ is $2m + 2 \leq r$. At every grid point, f is nonpositive. Positivity and exactness therefore imply

$$0 \geq \sum_i w_i f(i/N) = \int_0^1 f(x) dx = A_m/N - B_m.$$

This yields the first bound. For even r , $m = r/2 - 1$; for odd r , $m = (r - 3)/2$. Direct substitution gives the second bound in both cases. $\square$

Here is the promised verification of the Jacobi identities. Set $J_k^{(b)}(x) = P_k^{(0,b)}(2x - 1)$. Orthogonality gives

$$\int_0^1 x^b J_k^{(b)}(x) J_\ell^{(b)}(x) dx = \frac{\mathbf{1}_{k=\ell}}{2k + b + 1}.$$

The Jacobi contiguous relation, solved recursively, gives

$$J_m^{(3)} = \frac{1}{D_m} \sum_{k=0}^m (-1)^{m-k} (2k + 3)(k + 1)(k + 2) J_k^{(2)},$$

$$J_k^{(2)} = \frac{1}{(k + 1)(k + 2)} \sum_{j=0}^k (-1)^{k-j} 2(j + 1)^2 J_j^{(1)}, \quad D_m = (m + 1)(m + 2)(m + 3).$$

Thus

$$B_m = \frac{1}{D_m^2} \sum_{k=0}^m (2k + 3)(k + 1)^2 (k + 2)^2 = \frac{1}{3};$$

the sum telescopes because its summand is

$$\frac{(k+1)^2(k+2)^2(k+3)^2 - k^2(k+1)^2(k+2)^2}{3}.$$

Combining the two connection formulas yields

$$J_m^{(3)} = \frac{2}{D_m} \sum_{j=0}^m (-1)^{m-j} (j+1)^2 ((m+2)^2 - (j+1)^2) J_j^{(1)}.$$

Consequently

$$A_m = \frac{2}{D_m^2} \sum_{\ell=1}^{m+1} \ell^3 ((m+2)^2 - \ell^2)^2 = \frac{m^2 + 4m + 6}{12},$$

where the last equality follows by summing the powers ℓ^3, ℓ^5, ℓ^7 (or induction in m). The connection formulas follow directly from

$$(2k+b+1)J_k^{(b)} = (k+b+1)J_k^{(b+1)} + kJ_{k-1}^{(b+1)}.$$

Standard normalization and the contiguous relation are recorded at <https://dlmf.nist.gov/18.3> and <https://dlmf.nist.gov/18.9.E5>.

Lemma 2 (Necessity of the insertion criterion). *Let s be a permutation-fair string on r letters, with arbitrary multiplicities $c_1, \dots, c_r$. Form T from N unchanged copies of s , and insert runs of the new letter in gaps $i = 0, \dots, N$, with nonnegative integer lengths k_i and $K = \sum_i k_i > 0$. Then T is permutation-fair if and only if $w_i = k_i/K$ gives a degree- r uniform quadrature on the grid i/N .*

Proof. The partial-pattern formula for a fair string, and its powers, is

$$C_\pi(s^a) = \frac{a^{|\pi|}}{|\pi|!} \prod_{\ell \in \pi} c_\ell.$$

For a fixed full pattern of T placing the new letter after j old letters, its count therefore equals

$$\frac{KN^r \prod_{\ell=1}^r c_\ell}{r!} \sum_{i=0}^N w_i \binom{r}{j} (i/N)^j (1 - i/N)^{r-j}.$$

Its required fair count is $KN^r \prod_\ell c_\ell / (r+1)!$. Thus fairness is equivalent to the degree- r Bernstein moment equations. Bernstein polynomials span all polynomials of degree at most r , proving the claim. $\square$

Corollary 3 (An intrinsic $n \log n$ insertion cost). *Consider any sequence of constructions that starts from a one-letter string and adds one new letter at a time by placing runs of it between whole unchanged copies of the preceding string. Suppose every intermediate string is permutation-fair. The oldest die in the final n -letter string has at least*

$$\prod_{r=2}^{n-1} \frac{r^2}{16} = \frac{((n-1)!)^2}{16^{n-2}} = \exp(2n \log n - O(n))$$

faces (the first displayed bound is merely weak when a factor is below 1). In particular, no choice of quadrature weights or improved denominator estimate within this framework can give an $\exp(O(n))$ construction.

Proof. The r -to- $(r+1)$ insertion uses $N_r \geq r^2/16$ copies by the preceding results. Every such insertion multiplies the oldest die's face count by N_r . Multiply these inequalities and use Stirling's formula. $\square$

The same argument allows runs of zero length, hence omitted grid nodes, and any grouping of the unchanged copies into larger blocks. It also allows starting from an arbitrary fixed-size fair seed. It does not apply when the old string is rearranged or locally changed at an insertion, or when several letters are inserted jointly by a different mechanism.